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(54) **GUIDE SYSTEM FOR GUIDING AT LEAST ONE MOVABLY MOUNTED DOOR LEAF**

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(57) **ABSTRACT**

A guide system for guiding a movably-supported door wing, includes a carrier for pivotally supporting the door wing, a guide, in particular a guide rail, configured to be fixed to the furniture wall for moving the carrier along the furniture wall, a guide device for displaceably supporting the carrier along the at least one guide, a toothed rack, and a gear engaged or configured to be engaged in the toothed rack. The gear is movable at least partially along the toothed rack upon a movement of the carrier along the guide and being configured to be set into rotation by an engagement in the toothed rack, and the carrier includes a bearing device for supporting the toothed rack.

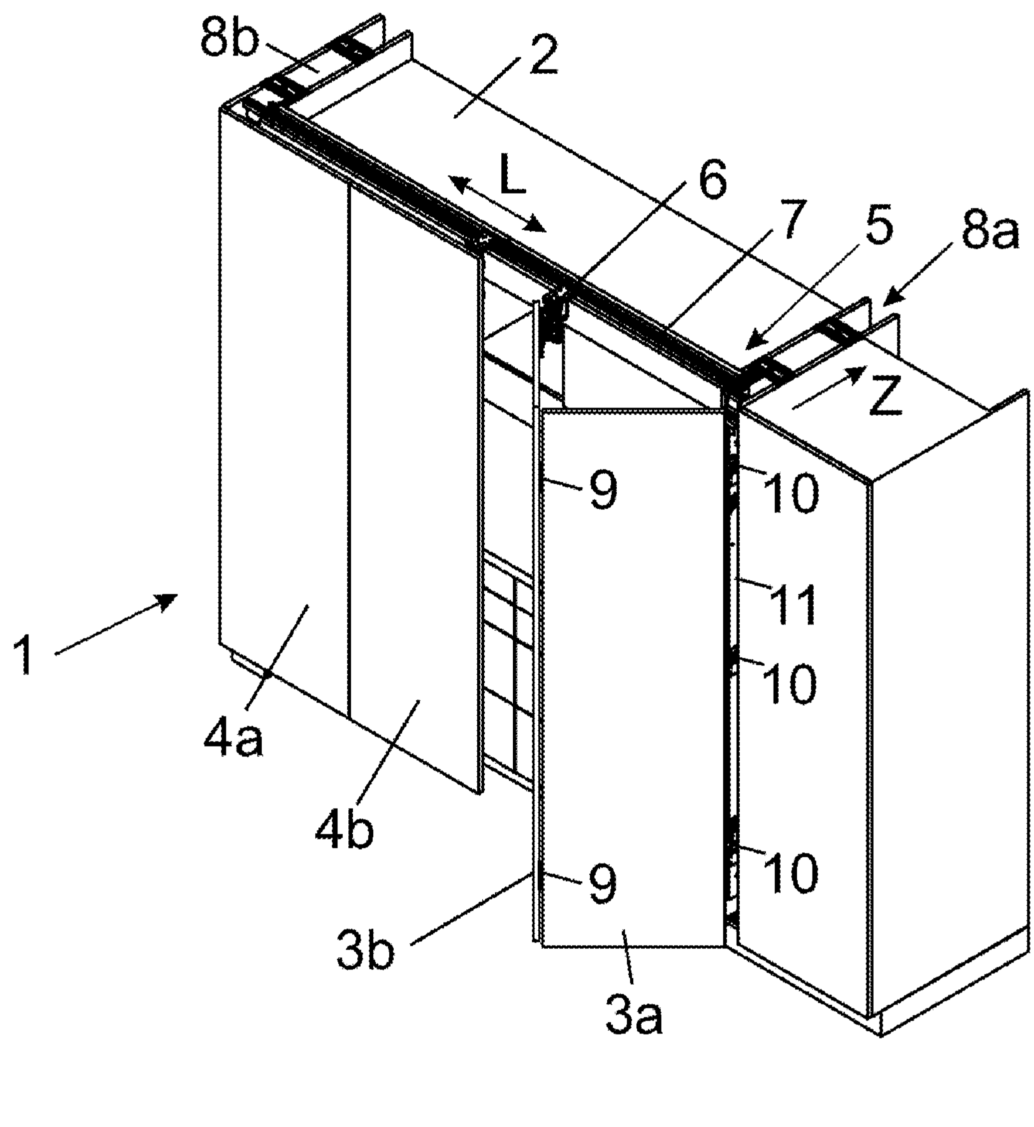


Fig. 1a

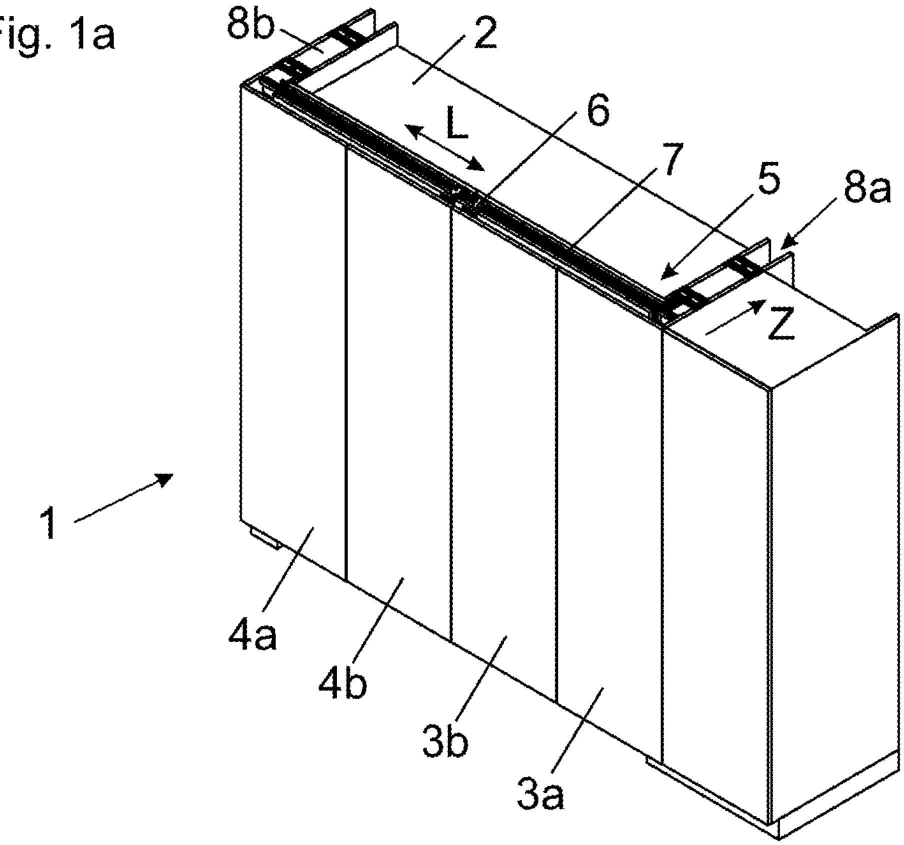


Fig. 1b

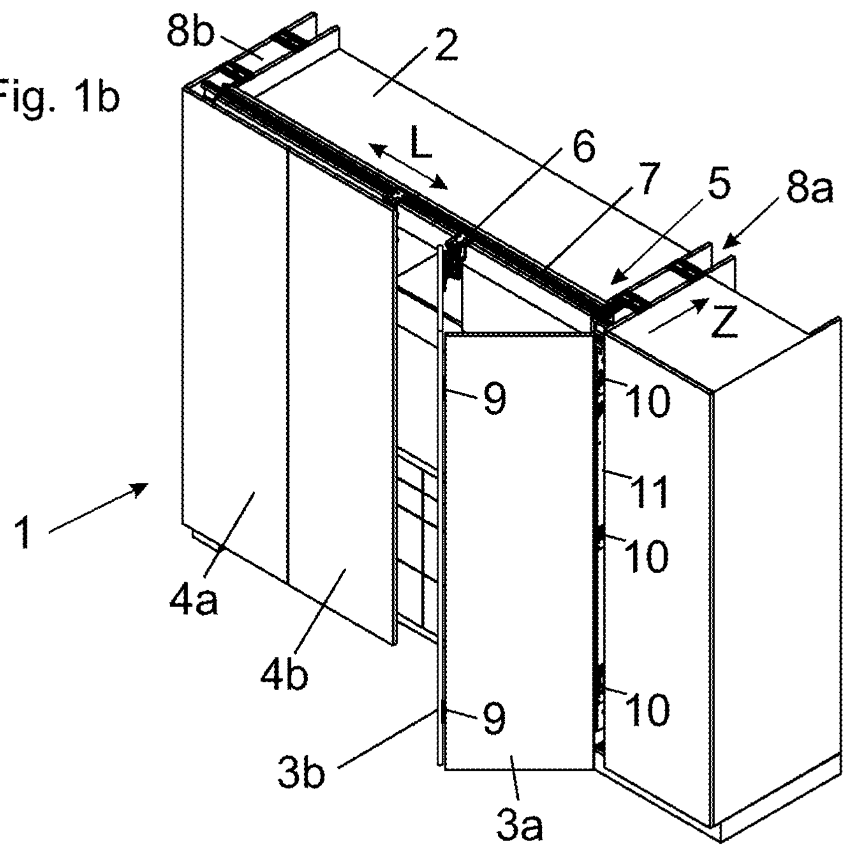


Fig. 2a

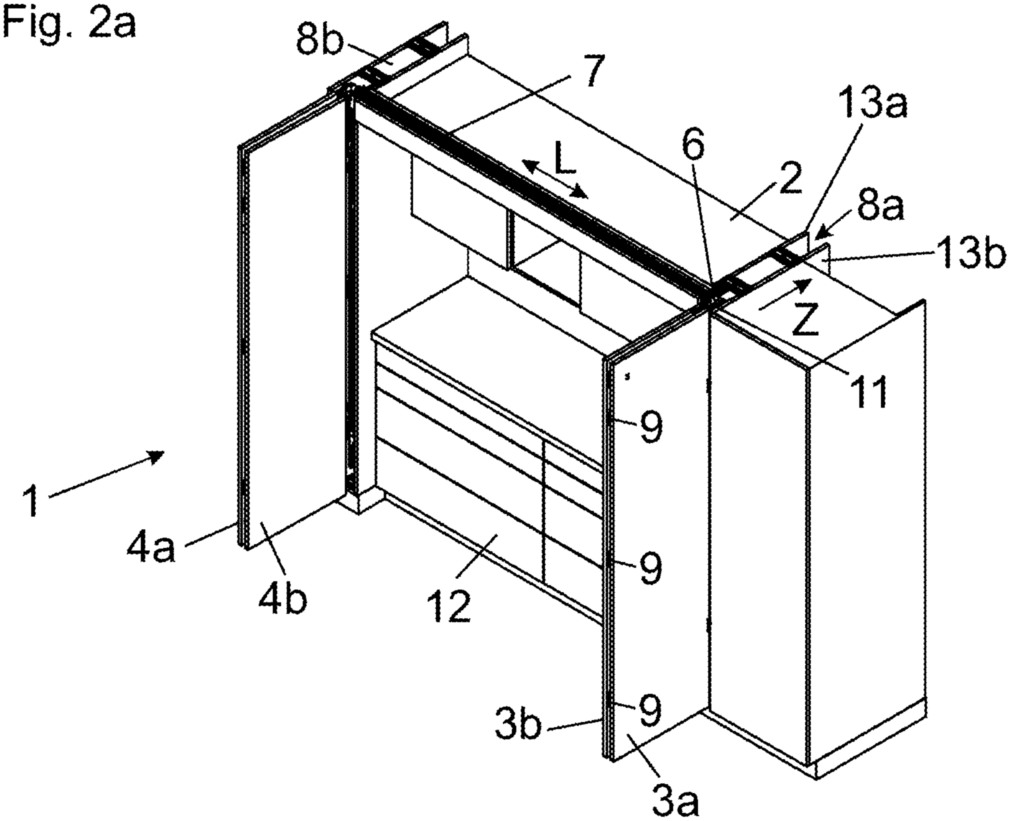
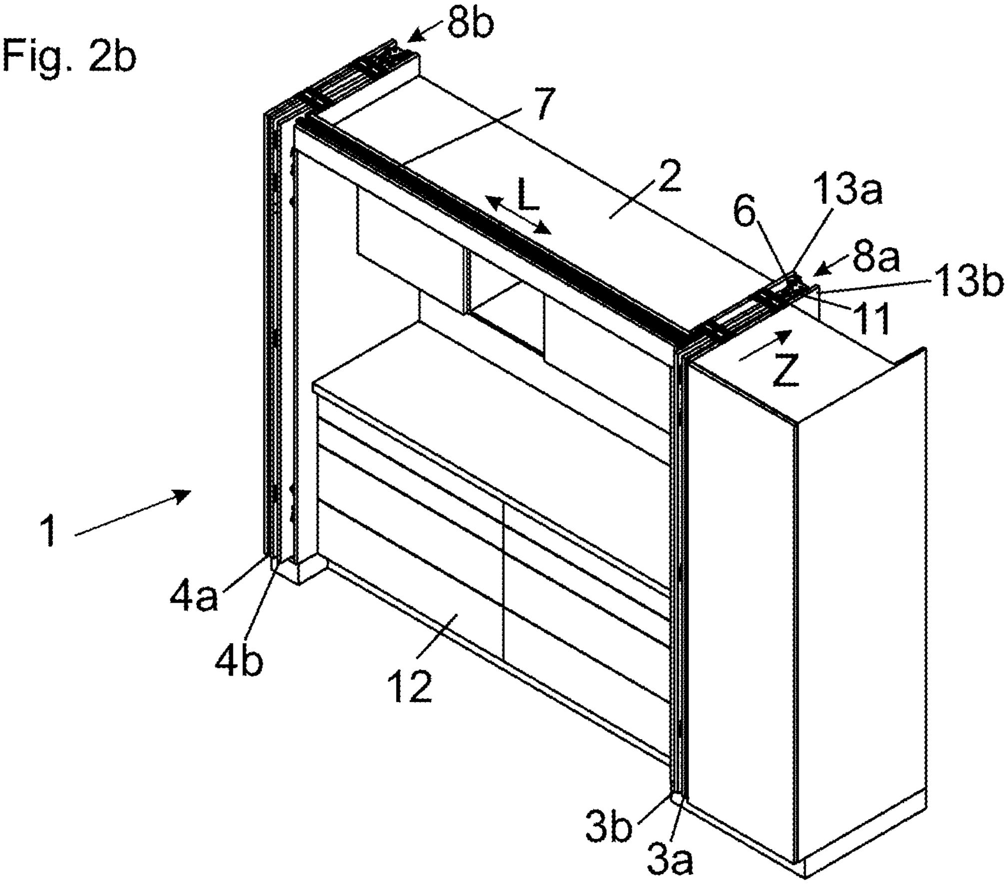

8b
7
L
6
2
13a
8a
13b
Z
11
9
9
9
1
4a
4b
12
3b
3a

Fig. 2b


8b
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L
2
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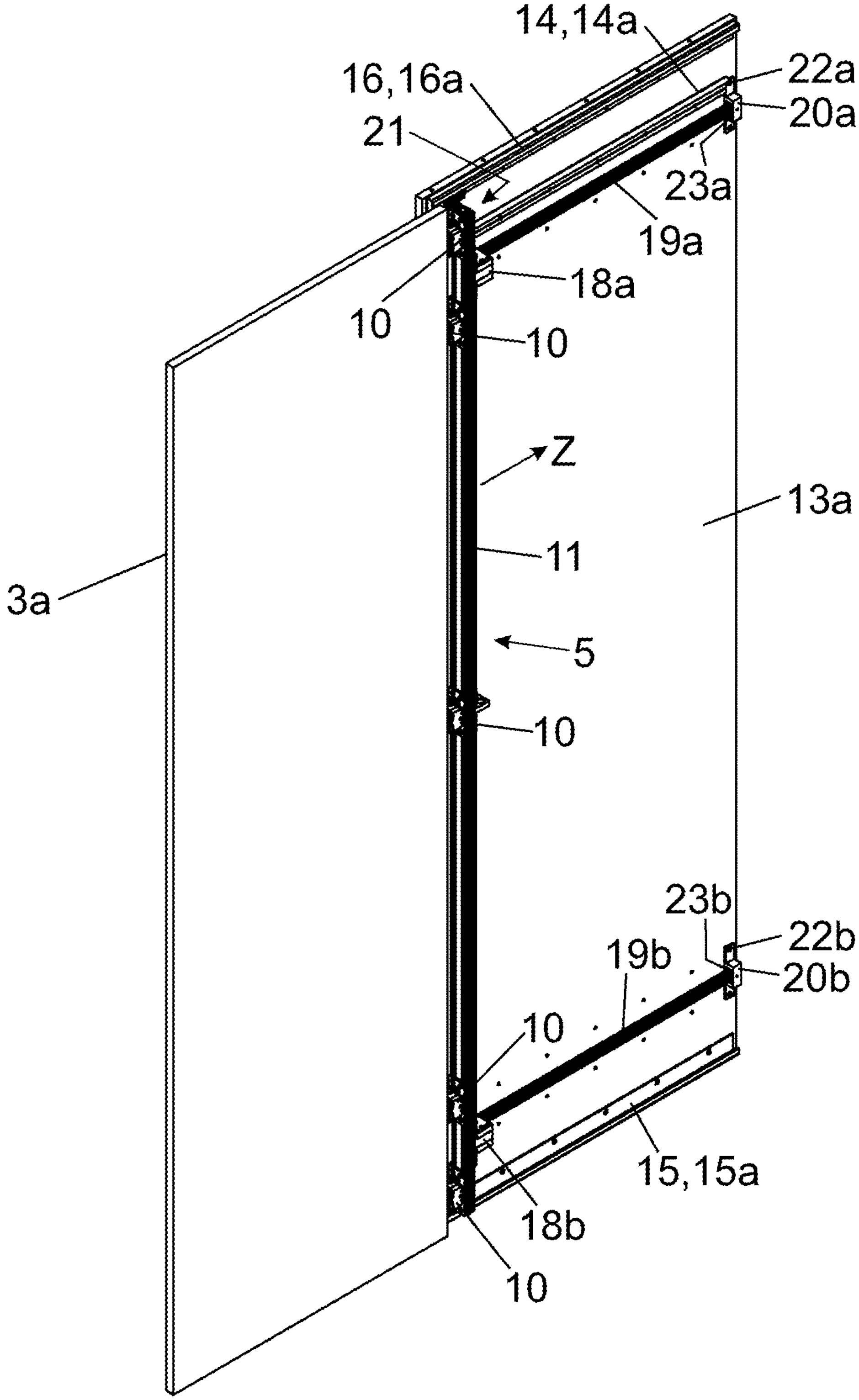
6
8a
13b
11
Z
1
4a
4b
12
3b
3a

Fig. 3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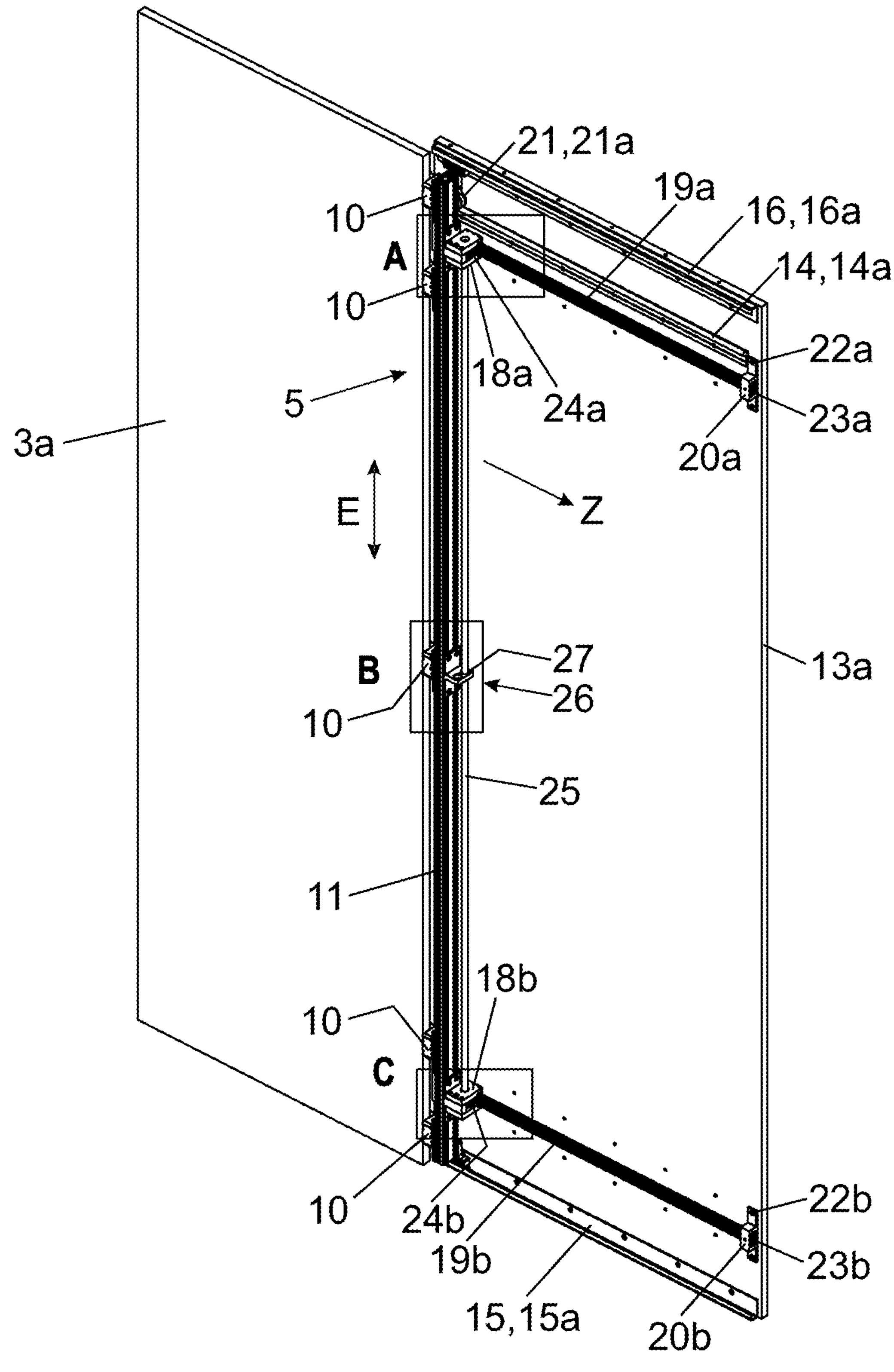
22a
16,16a
20a
21
23a
19a
18a
10
10
Z
13a
11
3a
5
10
23b
22b
19b
20b
10
15,15a
18b
10

Fig. 4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10
A
14,14a
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22a
18a
24a
23a
5
3a
20a
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Z
27
B
26
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11
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10
C
22b
10
24b
19b
23b
15,15a
20b

Fig. 5a

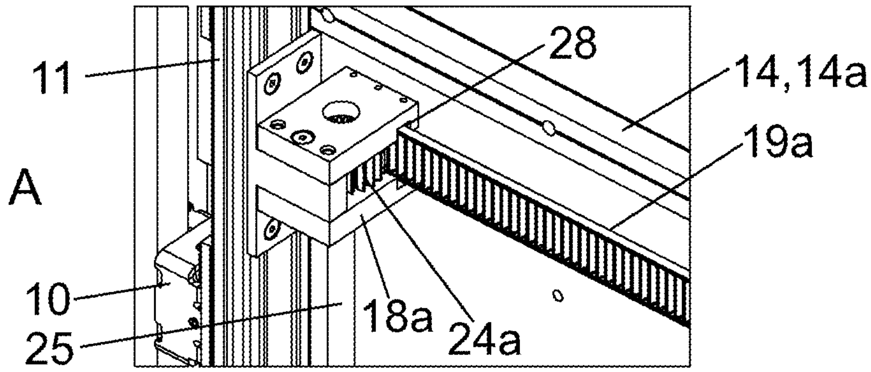


Fig. 5b

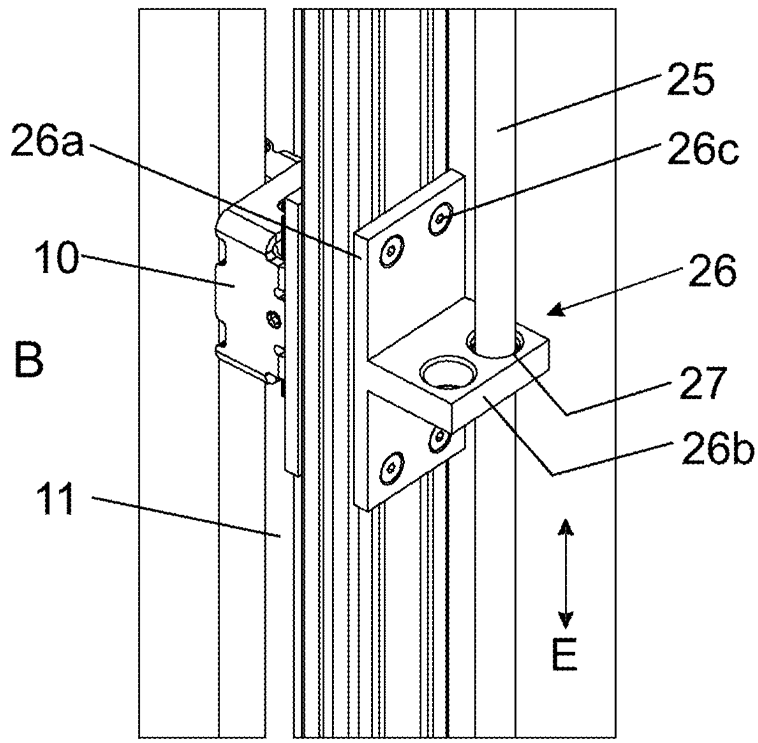


Fig. 5c

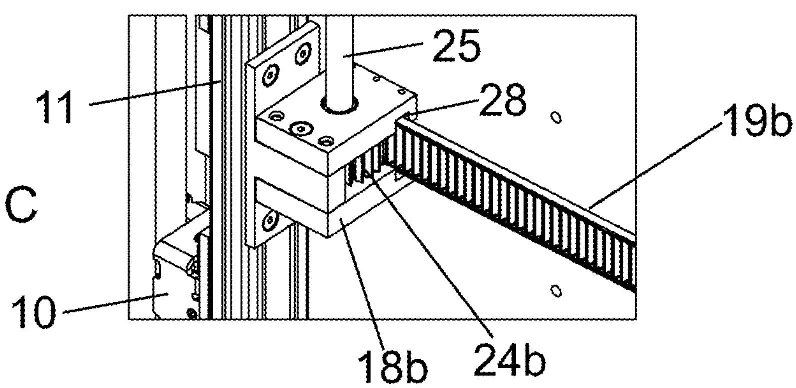
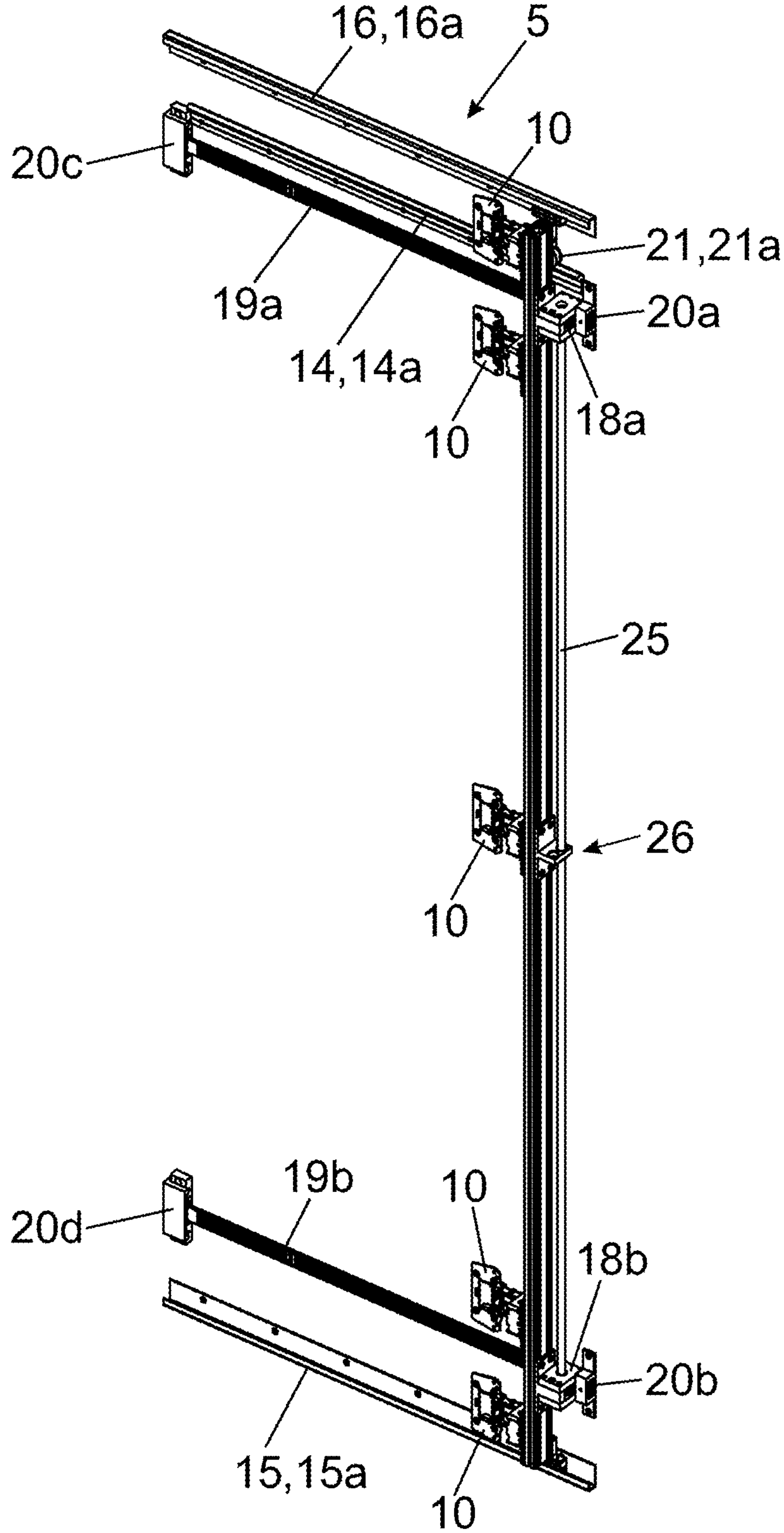


Fig. 6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10
20c
21,21a
20a
19a
14,14a
18a
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26
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19b
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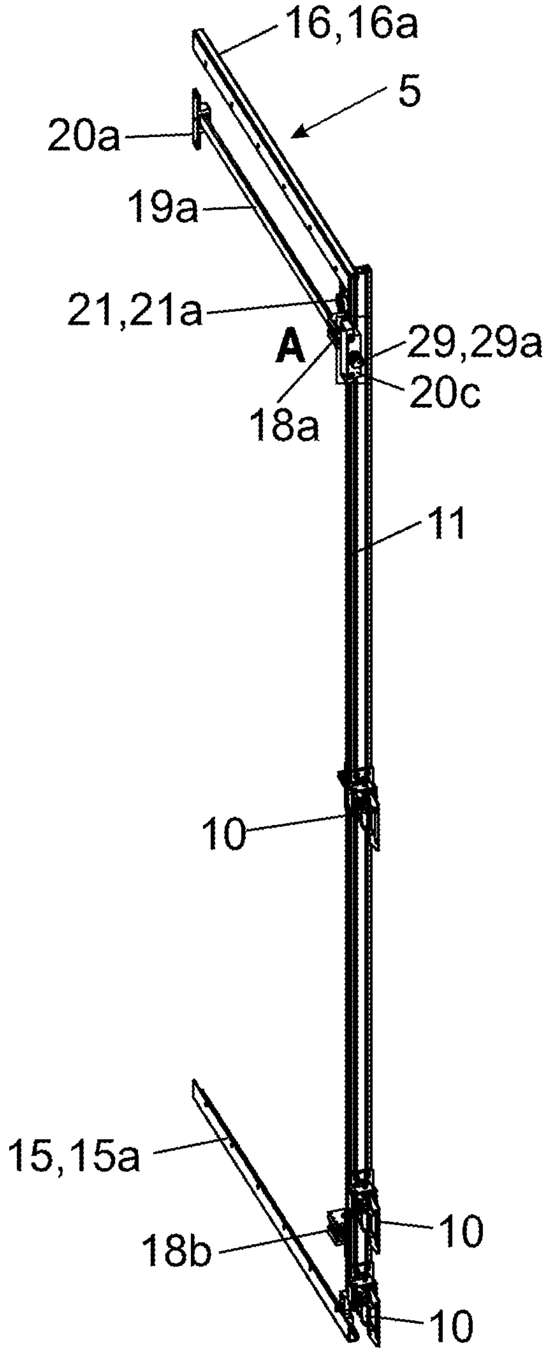
18b
20b
15,15a
10

Fig. 7a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20a
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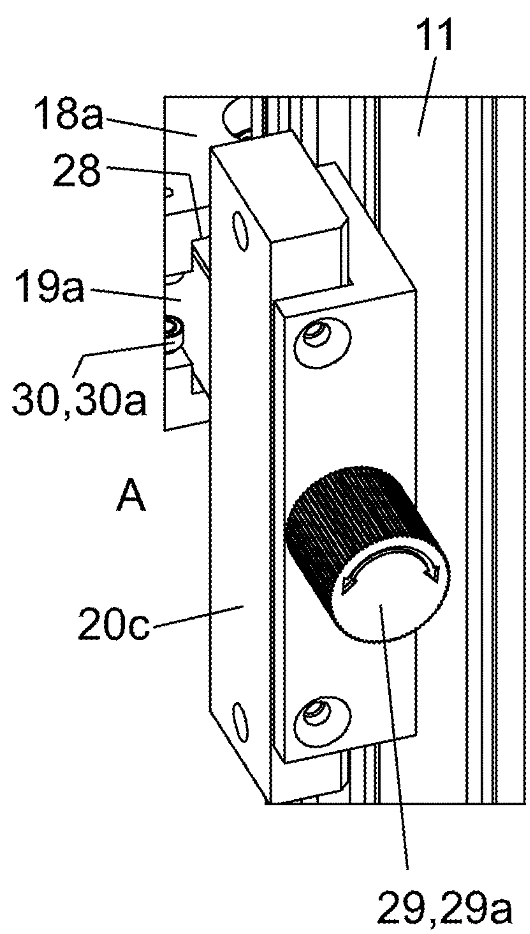
A
29,29a
20c
18a
11
10
15,15a
18b
10
10

Fig. 7b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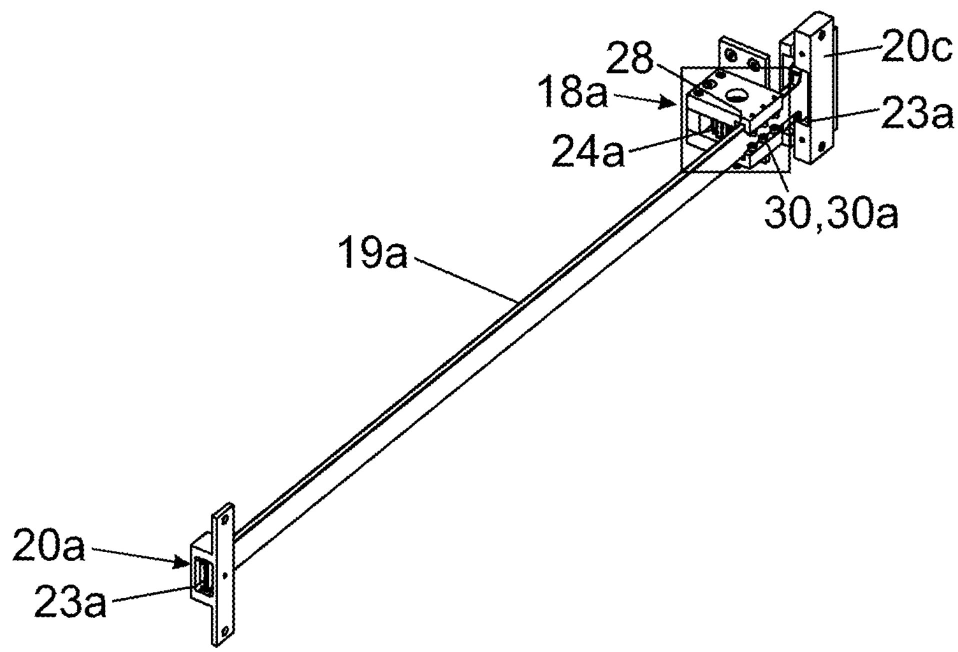
18a
28
19a
30,30a
A
20c
29,29a

Fig. 8a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18a
24a
20c
23a
30,30a
19a
20a
23a

Fig. 8b

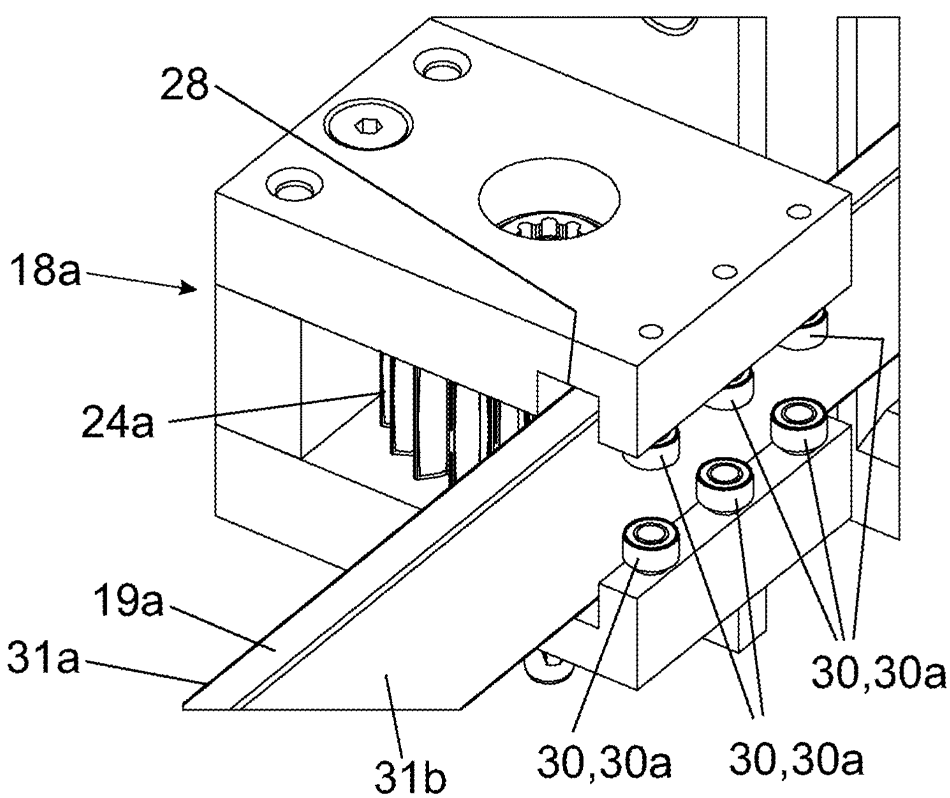
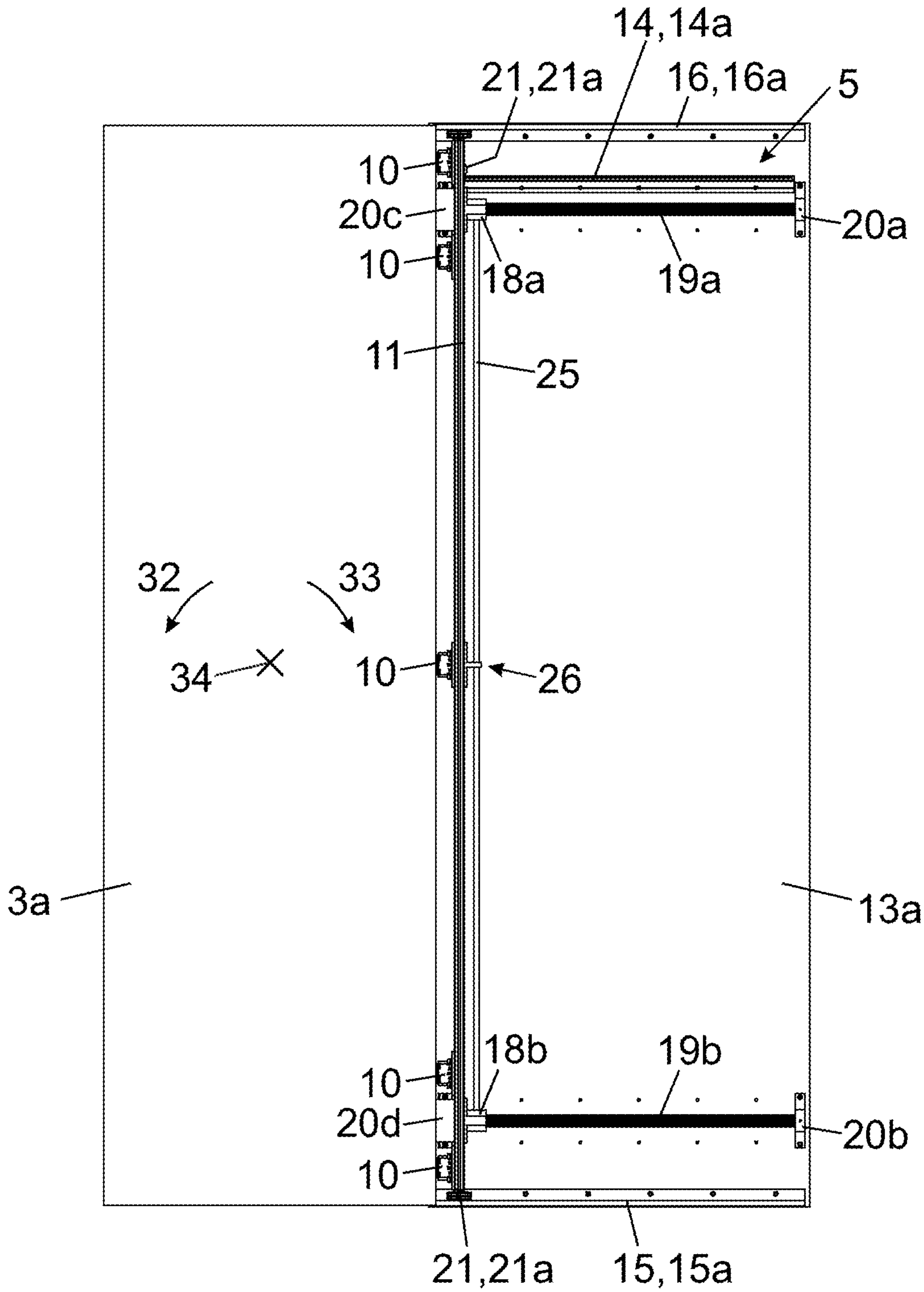

28
18a
24a
19a
31a
31b
30,30a
30,30a
30,30a

Fig. 9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20c
20a
10
18a
19a
11
25
32
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34
10
26
3a
13a
18b
19b
10
20d
20b
10
21,21a
15,15a

Fig. 10a

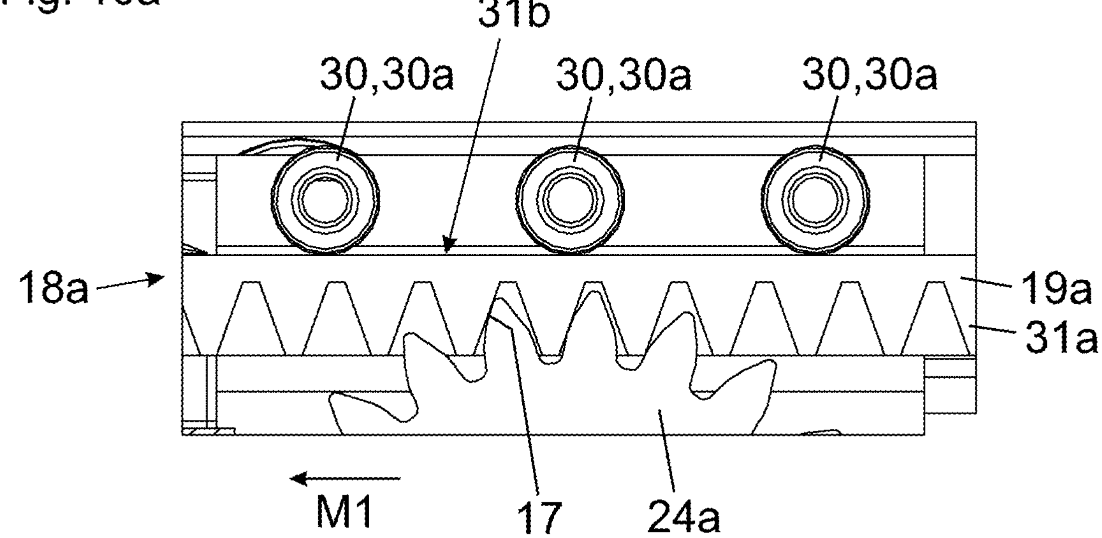
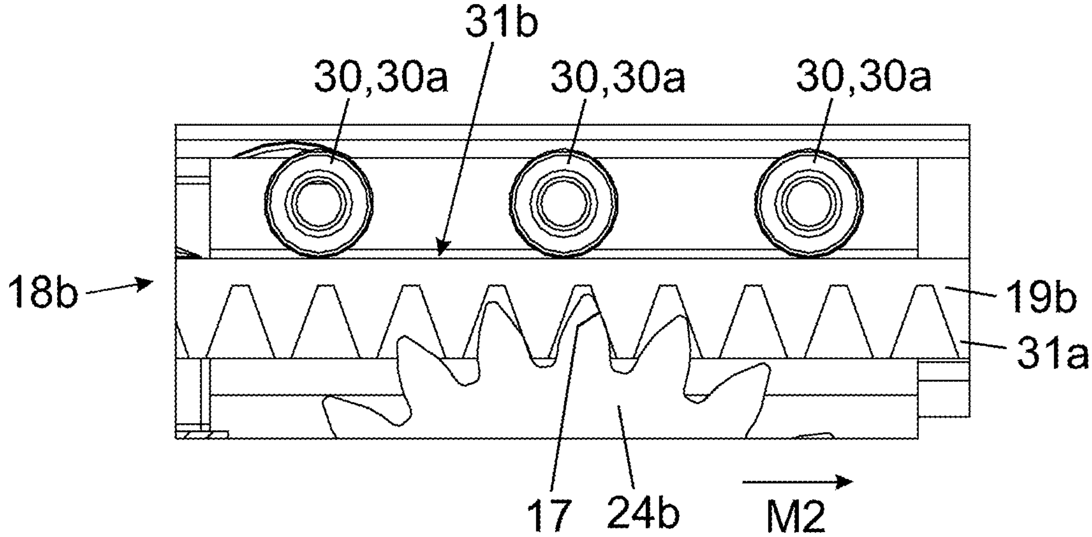


Fig. 10b



GUIDE SYSTEM FOR GUIDING AT LEAST ONE MOVABLY MOUNTED DOOR LEAF

[0001] The present application is a continuation of International Application PCT/AT2023/060382 filed on Nov. 10, 2023. Thus, all of the subject matter of International Application PCT/AT2023/060382 is incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to a guide system for guiding at least one movably-supported door wing, in particular a folding-sliding-door, along a furniture wall, the guide system comprising:

- **[0003]** at least one carrier for pivotally supporting the at least one door wing,
- **[0004]** at least one guide, in particular a guide rail, configured to be fixed to the furniture wall for moving the at least one carrier along the furniture wall,
- **[0005]** at least one guide device for displaceably supporting the at least one carrier along the at least one guide, and
- **[0006]** at least one toothed rack and at least one gear engaged or configured to be engaged in the toothed rack, the gear being movable at least partially along the toothed rack upon a movement of the carrier along the guide and being configured to be set into rotation by an engagement in the toothed rack.

[0007] Moreover, the invention concerns an item of furniture comprising a furniture carcass, at least one door wing, in particular a folding-sliding-door, movably supported relative to the furniture carcass, and at least one guide system of the type to be described for moving the at least one door wing relative to the furniture carcass.

[0008] US 2004/046488 A1 discloses a guide system for a door wing which is movable along a sidewall of the furniture carcass by the guide system. The door wing is thus insertable in a depth direction of the item of furniture, and the door wing reveals access to the interior of the furniture carcass in the inserted condition. The door wing is pivotally supported on a carrier via hinges. The carrier is movable along the sidewall of the furniture carcass via a plurality of toothed racks and via a plurality of gears engaging in the toothed racks, the toothed racks being spaced apart from each other in a height direction. By a synchronization shaft, the rotational movements of the gears can be synchronized with each other. A drawback of this construction is that the guide system is only suitable for door wings having a low weight. The challenge when guiding a door wing along a furniture wall is, in fact, that the door wing has a tendency to tilt due to its weight. This, in turn, leads to an undesired torsional movement of the carrier and to a jamming of the door wing relative to the guide. Moreover, it is possible that the upper gears no longer duly engage in their associated upper toothed racks, whereas the lower gears can get jammed with the lower toothed rack due to the occurring weight of the door wing. These circumstances lead to the fact that a movement of the door wing is made impossible or is at least substantially impeded. Moreover, undesired operating noises can occur, especially because all toothed racks are flatly fixed to the sidewall. In this way, it is possible that the furniture carcass becomes a resonator body due to the occurring oscillations, and the noises are additionally increased therewith.

SUMMARY OF THE INVENTION

[0009] It is an object of the present invention to propose a guide system of the type mentioned in the introductory part, thereby avoiding the drawbacks as discussed above.

[0010] According to the invention, the at least one carrier includes at least one bearing device for supporting the at least one toothed rack.

[0011] In other words, the at least one toothed rack is supported on the carrier on which the at least one door wing is to be pivotally arranged, the carrier being movable in the mounted position along at least one guide arranged on the furniture wall. In this way, it can be ensured that the at least one gear has a defined spacing with respect to the at least one toothed rack, and the gear is always movable relative to the toothed rack with a predefined engagement depth. Moreover, disturbing operating noises can be thereby reduced.

[0012] According to an embodiment, the at least one bearing device and the at least one toothed rack are movable relative to each other, preferably displaceable relative to each other.

[0013] According to a preferred embodiment, the at least one bearing device includes at least one linear guide, preferably a U-shaped profile, for movably supporting, preferably displaceably supporting, the at least one toothed rack.

[0014] Due to the at least one linear guide, the engagement depth of the at least one gear can be limited such that the teeth of the at least one gear only partially engage in the teeth spaces of the at least one toothed rack. In this way, the engagement depth of the at least one gear can be limited such that a jamming or a disengagement of the at least one gear with respect to the at least one toothed rack is prevented.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] Further details and advantages of the present invention will be apparent from the following description of figures, in which:

[0016] FIG. **1***a*, **1***b* are a perspective view of an item of furniture comprising a furniture carcass and door wings movable relative thereto,

[0017] FIG. **2***a*, **2***b* show the item of furniture according to FIGS. **1***a*, **1***b* in further positions of the door wings relative to each other,

[0018] FIG. **3** is a perspective view of a furniture wall with the guide system,

[0019] FIG. **4** is a further perspective view of a furniture wall with the guide system,

[0020] FIG. **5***a*-**5***c* are enlarged detail views according to FIG. **4**,

[0021] FIG. **6** shows the guide system in a perspective view,

[0022] FIG. **7***a*, **7***b* show the guide system in a perspective view from the front and an enlarged detail view thereof,

[0023] FIG. **8***a*, **8***b* are a perspective view of a bearing device with a toothed rack supported thereon, and an enlarged detail view thereof,

[0024] FIG. **9** is a side view of the furniture wall with a door wing fully extended, and

[0025] FIG. **10***a*, **10***b* shows the (upper) first bearing device and the (lower) second bearing device in engagement with the (upper) first toothed rack and the (lower) second toothed rack.

DETAILED DESCRIPTION OF THE INVENTION

[0026] FIG. **1***a* shows a perspective view of an item of furniture **1** comprising a furniture carcass **2** and door wings **3***a*, **3***b*; **4***a*, **4***b* movable relative to the furniture carcass **2**. The door wings **3***a*, **3***b* and the door wings **4***a*, **4***b* are movable by a guide system **5** between a first position, in which the door wings **3***a*, **3***b*; **4***a*, **4***b* are aligned substantially coplanar to each other, and a second position, in which the door wings **3***a*, **3***b*; **4***a*, **4***b* are aligned substantially parallel to each other.

[0027] The door wings **3***a*, **3***b* are insertable into a lateral receiving compartment **8***a* of the furniture carcass **2** when located in the second (parallel) position, whereas the door wings **4***a*, **4***b* are insertable into a further receiving compartment **8***b* when located in a parallel position to each other.

[0028] The functionality will be explained in the following with the aid of the door wings **3***a* and **3***b*, and the same explanations are valid for the door wings **4***a*, **4***b*.

[0029] The guide system **5** includes a guide rail **7** having a longitudinal direction (L), and a guide carriage **6** configured to be connected to the second door wing **3***b* is displaceably supported along the guide rail **7**. In a mounted condition, the guide rail **7** extends substantially parallel to the front face of the furniture carcass **2**.

[0030] FIG. **1***b* shows the item of furniture **1**, in which the door wings **3***a*, **3***b* have been moved from the coplanar position shown in FIG. **1***a* into an angled position relative to each other. The first door wing **3***a* is pivotally supported on a carrier **11** via two or more hinges **10**, and the carrier **11** (jointly with the door wings **3***a*, **3***b*) is insertable in a depth direction (Z) into the lateral receiving compartment **8***a*.

[0031] In FIG. **1***b*, the carrier **11** is located in a transfer position in which the carrier **11** adjoins the guide rail **7** in the longitudinal direction (L) such that the guide carriage **6** is transferable to and from between the guide rail **7** and the carrier **11**. In the shown transfer position, the carrier **11** is releasably locked to the guide rail **7**, and the locking between the guide rail **7** and the carrier **11** is releasable by an entry of the running carriage **6** in or onto the carrier **11**.

[0032] The carrier **11** is in the form of a longitudinally extending column, and the length of which is at least half of a height of the door wings **3***a*, **3***b*. The two door wings **3***a*, **3***b* are hingedly connected to each other about a vertically extending axis via at least one hinge fitting **9**. The second door wing **3***b* is displaceably supported along the guide rail **7** via the guide carriage **6**.

[0033] FIG. **2***a* shows the item of furniture **1** with the door wings **3***a* and **3***b* which are now aligned parallel to each other. The carrier **11** has been unlocked from the guide rail **7** by an entry of the guide carriage **6**, and the carrier **11** (jointly with the guide carriage **6** and the door wings **3***a*, **3***b*) is insertable in a depth direction (Z) of the furniture carcass **2** into the receiving compartment **8***a*. The depth direction (Z) extends transversely, preferably substantially perpendicular, to the longitudinal direction (L) of the guide rail **7**.

[0034] FIG. **2***b* shows the item of furniture **1** with the door wings **3***a*, **3***b* which are now located in a fully inserted condition within the receiving compartment **8***a*. The door wings **3***a*, **3***b* are thus movable by the guide system **5** between a first position according to FIG. **1***a*, in which the door wings **3***a*, **3***b* are aligned substantially coplanar to each other, and a second position according to FIG. **2***b*, in which the door wings **3***a*, **3***b* are aligned substantially parallel to each other and being received within the receiving compartment **8***a*.

[0035] In this way, for example a kitchen **12**, as shown in FIG. **2***a*, **2***b*, can be entirely covered so as to separate the kitchen **12** from a remaining area of a living room.

[0036] In the shown embodiment, the receiving compartment **8***a* is formed by a first furniture wall **13***a* and by a second furniture wall **13***b* spaced apart from the first furniture wall **13***a* in a substantially parallel relationship. The door wings **3***a*, **3***b*, in a parallel position to each other, are insertable between the two furniture walls **13***a*, **13***b*. In the shown embodiment, each of the two furniture walls **13***a*, **13***b* are configured as sidewalls of the furniture carcass **2**.

[0037] FIG. **3** shows a perspective view of the furniture wall **13***a* with the guide system **5** for guiding the at least one door wing **3***a*. For the sake of improved overview, the second door wing **3***b* hingedly connected to the first door wing **3***a* is not illustrated.

[0038] The guide system **5** includes at least one longitudinally extending carrier **11** on which the at least one door wing **3***a* can be pivotally supported, for example via two or more hinges **10**. The carrier **11** can have a length corresponding substantially to a height of the door wing **3***a*.

[0039] The guide system **5** includes at least one guide **14**, preferably a guide rail **14***a*, configured to be fixed to the furniture wall **13***a*, the guide **14** being provided for moving the at least one carrier **11** along the furniture wall **13***a*.

[0040] In the shown embodiment, further guides **15**, **16**, preferably further guide rails **15***a*, **16***a*, are provided by which the at least one carrier **11** is displaceable along the furniture wall **13***a*. The guides **14**, **15**, **16** are fixed to the furniture wall **13***a* with a vertical distance from each other.

[0041] The carrier **11** is displaceable along the at least one guide **14** via at least one guide device **21**. The guide device **21** can include at least one running wheel **21***a* (FIG. **4**), preferably a plurality of running wheels **21***a*, configured to run along the at least one guide **14**.

[0042] The guide device **21** can include a plurality of running wheels **21***a* rotationally supported about a rotational axis oriented in a first direction. Moreover, further running wheels rotationally supported about a rotational axis oriented in a second direction can be provided.

[0043] The guide system **5** includes at least one toothed rack **19***a*, and at least one gear **24***a* (FIG. **4**) is displaceable on the at least one toothed rack **19***a* upon a movement of the carrier **11** along the at least one guide **14**.

[0044] In the shown embodiment, at least one second toothed rack **19***b* is provided, and at least one second gear **24***b* (FIG. **4**) is displaceable along the second toothed rack **19***b* upon a movement of the carrier **11** along the guides **14**, **15**, **16**.

[0045] According to the invention, the carrier **11** includes at least one bearing device **18***a*, **18***b* for supporting the at least one toothed rack **19***a*, **19***b*.

[0046] The toothed rack **19***a*, **19***b* has an end section, and at least one holding device **20***a*, **20***b* configured to be fixed to the furniture wall **13***a* is provided for supporting the end section of the toothed rack **19***a*, **19***b*.

[0047] The holding device **20***a*, **20***b* includes at least one fastening location **22***a*, **22***b* for fixing an end section of the toothed rack **19***a*, **19***b* to the furniture wall **13***a*. For example, the fastening location **22***a*, **22***b* of the holding device **20***a*,

20*b* can include a hole for the passage of a screw and/or at least one dowel to be fixed to the furniture wall **13***a*.

[0048] Each of the holding devices **20***a*, **20***b* includes a guide **23***a*, **23***b* for movably supporting the toothed rack **19***a*, **19***b*. In other words, the toothed rack **19***a*, **19***b* is displaceably supported relative to the holding device **20***a*, **20***b*.

[0049] For example, the at least one gear **24***a*, **24***b* (FIG. **4**) configured to run along the toothed rack **19***a*, **19***b* can be rotationally arranged on the carrier **11** or on the at least one bearing device **18***a*, **18***b* of the carrier **11**.

[0050] FIG. **4** shows a further perspective view of the furniture wall **13***a* with the guide system **5** according to FIG. **3**. The two bearing devices **18***a*, **18***b* of the carrier **11** can be seen, and at least one gear **24***a*, **24***b* is rotationally supported on each or in each of the bearing devices **18***a*, **18***b* or on the carrier **11**. The at least one gear **24***a*, **24***b* can be rotationally supported about an axis extending parallel to the longitudinal direction (E) of the carrier **11**, and is configured so as to mesh with the at least one toothed rack **19***a*, **19***b*.

[0051] For synchronizing a rotational movement of the gears **24***a*, **24***b*, at least one synchronization shaft **25** is provided, the synchronization shaft **25** extending substantially parallel to the longitudinal direction (E) of the carrier **11**. As a result, a rotational movement of the gears **24***a*, **24***b* can be synchronized with each other by a rotational movement of the synchronization shaft **25** about its own axis.

[0052] In the shown embodiment, at least one stabilizing device **26** is provided, the stabilizing device **26** being configured to prevent or at least to limit a bulging movement of the synchronization shaft **25** in a direction extending transversely to the longitudinal direction (E) of the synchronization shaft **25**.

[0053] The at least one stabilizing device **26** can be arranged in a central region of the synchronization shaft **25** and/or can include at least one opening **27** for the passage of the synchronization shaft **25**.

[0054] FIGS. **5***a*-**5***c* each show an enlarged detail view of FIG. **4**.

[0055] FIG. **5***a* shows the framed region "A" according to FIG. **4** in an enlarged view. To be seen is the bearing device **18***a* for movably supporting the toothed rack **19***a*, the bearing device **18***a* being fixed to the carrier **11**. The bearing device **18***a* includes at least one rotatable gear **24***a* meshing with the toothed rack **19***a*.

[0056] The bearing device **18***a* can include at least one linear guide **28** for movably supporting the toothed rack **19***a*. The bearing device **18***a* is movably supported relative to the toothed rack **19***a* upon a movement of the carrier **11** along the guide **14**. In the shown embodiment, the at least one guide **28** includes a U-shaped profile in which the toothed rack **19***a* can be guided.

[0057] FIG. **5***b* shows the framed region "B" according to FIG. **4** in an enlarged view. The stabilizing device **26** can be seen, the stabilizing device **26** being configured to prevent or at least to limit a bulging movement of the synchronization shaft **25** in a direction extending transversely to the longitudinal direction (E) of the synchronization shaft **25**.

[0058] In the shown embodiment, the stabilizing device **26** includes a fastening section **26***a* configured to bear against the carrier **11**, and further includes a holding limb **26***b* for holding the synchronization shaft **25**. The holding limb **26***b* projects transversely, preferably perpendicular, from the fastening section **26***a*. The fastening section **26***a* includes at least one fastening location **26***c* for fixing the stabilizing device **26** to the carrier **11**. The holding limb **26***b* includes at least one, for example cylindrical, opening **27** for the passage of the synchronization shaft **25**.

[0059] FIG. **5***c* shows the framed region "C" according to FIG. **4** in an enlarged view. The lower, second bearing device **18***b* includes at least one second gear **24***b* configured to run along the second toothed rack **19***b*. The rotational movements of the two gears **24***a*, **24***b* can be synchronized with each other via the at least one synchronization shaft **25**. The second bearing device **18***b* also includes a linear guide **28**, preferably a U-shaped profile, configured to receive and to guide the second toothed rack **19***b*.

[0060] FIG. **6** shows the guide system **5** in a further perspective view. In FIG. **6**, the guide system **5** is located in a position in which the door wing **3***a* is entirely located within the receiving compartment **8***a* (corresponding to the position according to FIG. **2***b*). For reasons of improved overview, the door wing **3***a* is not depicted.

[0061] Each of the toothed racks **19***a*, **19***b* includes a first end section and a second end section. The first end section of the toothed rack **19***a*, **19***b* is supported on a first holding device **20***a*, **20***b*, and the second end section of the toothed rack **19***a*, **19***b* is supported on a second holding device **20***c*, **20***d*. Each of the holding devices **20***a*, **20***b*, **20***c*, **20***d* is to be fixed to the furniture wall **13***a*, and the bearing device **18***a*, **18***b* is movable between the two holding devices **20***a*, **20***b*, **20***c*, **20***d*.

[0062] In other words, each of the end sections of the toothed racks **19***a*, **19***b* are supported on a holding device **20***a*, **20***b*, **20***c*, **20***d*, and each of the holding devices **20***a*, **20***b*, **20***c*, **20***d* is to be fixed to the furniture wall **13***a*.

[0063] Preferably, the holding devices **20***a*, **20***b*, **20***c*, **20***d* are configured to hold the at least one toothed rack **19***a*, **19***b* with a distance to the furniture wall **13***a* in a fixed condition of the holding device **20***a*, **20***b*, **20***c*, **20***d*.

[0064] The bearing device **18***a*, **18***b* is thus movable between the holding devices **20***a*, **20***b*, **20***c*, **20***d*, and the toothed rack **19***a*, **19***b* is spaced apart from the furniture wall **13***a*. As a result, the relative spacing between the gear **24***a*, **24***b* and the toothed rack **19***a*, **19***b* can be maintained in a constant manner, so that the gear **24***a*, **24***b* is always movable with a constant engagement depth along the toothed racks **19***a*, **19***b*.

[0065] FIG. **7***a* shows the guide system **5** in a perspective view from the front. The guides **15**, **16**, preferably in the form of guide rails **15***a*, **16***a*, can be seen, the carrier **11** with the hinges **10** for movably supporting the door wing **3***a* being movable along the guides **15**, **16** in a mounted position.

[0066] One end of the toothed rack **19***a* is to be fixed to the furniture wall **13***a* via the holding device **20***a*. The other end of the toothed rack **19***a* is also to be fixed to the furniture wall **13***a* via the holding device **20***c* (FIG. **6**). The bearing device **18***a* of the carrier **11** is movable along the toothed rack **19***a* between the two holding devices **20***a*, **20***c*.

[0067] For compensating for a malposition between the (upper) first toothed rack **19***a* and the (lower) second toothed rack **19***b*, at least one adjustment device **29** is provided. By the adjustment device **29**, a position of the toothed rack **19***a* relative to the at least one holding device **20***a*, **20***c* is adjustable.

[0068] For example, the adjustment device **29** can include at least one rotatable adjustment wheel **29***a*, and a position of the toothed rack **19***a* relative to the holding device **18***a* is adjustable by rotating the at least one adjustment wheel **29***a*.

[0069] As a result, a malposition, in particular a tilting movement, of the at least one door wing **3***a* can be corrected by the adjustment device **29**. The malposition is caused by a torsional movement of the carrier **11** and/or by a weight of the door wing **3***a*.

[0070] In other words, the position of the carrier **11** relative to the toothed rack **19***a* is adjustable by the adjustment device **29**, and a movability between the toothed rack **19***a* and the holding devices **20***a*, **20***c* to be fixed to the furniture wall **13***a* has to be permitted.

[0071] According to an embodiment, the front end of the toothed rack **19***a* is connected to the adjustment element **29***a* of the holding device **20***c* in a movement-coupled manner. In contrast, the rear end of the toothed rack **19***a* is glidingly guided in the guide **23***a* (FIG. **4**) of the holding device **20***a*.

[0072] FIG. **7***b* shows the framed region "A" of FIG. **7***a* in an enlarged view. The adjustment wheel **29***a* is rotationally supported on the frontal holding device **20***c*. By rotating the adjustment wheel **29***a*, a position of the toothed rack **19***a* relative to the holding devices **20***a*, **20***c* is adjustable. To be seen is the linear guide **28** of the bearing device **18***a* for movably supporting the toothed rack **19***a*.

[0073] At least one gliding device **30** can be provided, the gliding device **30** being configured to reduce a frictional resistance between the at least one bearing device **18***a*, **18***b* and the at least one toothed rack **19***a*, **19***b*. For example, the at least one gliding device **30** can include a rolling body **30***a* rotatable about an axis, preferably about a vertically extending axis in the mounted position.

[0074] Due to the tilting movement of the at least one door wing **3***a* caused by the weight, it is possible that a side of the toothed rack **19***a*, opposing the side having the provided teeth, is pressed against the bearing device **18***a*. For this reason, the gliding device **30** for reducing the frictional resistance between the toothed rack **19***a* and the bearing device **18***a* is provided.

[0075] The at least one adjustment wheel **29***a* of the adjustment device **29** can be configured for manual actuation and/or is directly accessible for a person from the front in a mounted condition of the guide system **5**.

[0076] FIG. **8***a* shows a perspective view of the toothed rack **19***a* with the two holding devices **20***a*, **20***c* to be fixed to the furniture wall **13***a*, and with the bearing device **18***a* for movably supporting along the toothed rack **19***a*.

[0077] Each of the holding devices **20***a*, **20***c* to be fixed to the furniture wall **13***a* includes a guide **23***a* for movably supporting the toothed rack **19***a*.

[0078] FIG. **8***b* shows the framed region of FIG. **8***a* in an enlarged view. At least one gear **24***a* is rotationally supported on the bearing device **18***a*, the gear **24***a* being movable along the toothed rack **19***a*.

[0079] The at least one toothed rack **19***a* has a first longitudinal side **31***a* having teeth which are in engagement or which can be brought in engagement with the at least one gear **24***a*. The toothed rack **19***a* further includes a second longitudinal side **31***b* opposing the first longitudinal side **31***a*, and the at least one gliding device **30** is configured to bear against the second longitudinal side **31***b* of the toothed rack **19***a*.

[0080] The gliding device **30** for reducing the frictional resistance between the toothed rack **19***a* and the bearing device **18***a* can include at least one rolling body **30***a* configured to bear against the second longitudinal side **31***b* of the toothed rack **19***a*.

[0081] In the shown embodiment, the gliding device **30** includes a plurality of rolling bodies **30***a* which are spaced apart from each other in a longitudinal direction of the toothed rack **19***a* and/or which are arranged in at least two running planes arranged above each other in a mounting position.

[0082] FIG. **9** shows a side view of the furniture wall **13***a* with the door wing **3***a* fully extended. The arrangement consisting of toothed racks **19***a*, **19***b*, gears **24***a*, **24***b*, bearing devices **18***a*, **18***b* and, if appropriate, the synchronization shaft **25**, collectively form a compensating device for compensating for a tilting moment **32** of the carrier **11** or of the door wing **3***a* arranged thereon about a tilting axis **34** by a return moment **33**.

[0083] When the carrier **11** or the door wing **3***a* is vertically aligned, the tilting axis **34** is horizontally aligned. The compensating device with the aforementioned components thus ensures that the at least one door wing **3***a* is stably guided in each position, thereby preventing an undue tilting movement of the door wing **3***a* relative to the guides **14**, **15**, **16**.

[0084] FIG. **10***a* shows the (upper) first bearing device **18***a* in engagement with the (upper) first toothed rack **19***a*. The gear **24***a* is rotationally supported on the bearing device **18***a*. The bearing device **18***a* is arranged on the carrier **11**, and the door wing **3***a* is pivotally supported on the carrier **11** via the hinges **10**. This means that the gear **24***a* is pulled in the direction M1 by the weight of the door wing **3***a*. The tooth flank **17** of the gear **24***a* is thus pressed against the teeth of the toothed rack **19***a* in the direction M1. Due to the co-operation of the oblique tooth flank **17** of the gear **24***a* with the oblique toothing of the toothed rack **19***a*, a resulting force presses the second longitudinal side **31***b* (see FIG. **8***b*), opposing the first longitudinal side **31***a*, against the gliding device **30** having the rolling bodies **30***a*. Accordingly, the gliding device **30** is thus provided to reduce the frictional resistance between the second longitudinal side **31***b* of the toothed rack **19***a* and the upper bearing device **18***a*.

[0085] FIG. **10***b*, in contrast, shows the (lower) second bearing device **18***b* in engagement with the (lower) second toothed rack **19***b*. The second gear **24***b* is rotationally supported on the second bearing device **18***b*. The second bearing device **18***b* is arranged on the carrier **11** so as to be spaced apart from the first bearing device **18***a*, and the door wing **3***a* is pivotally supported on the carrier **11** via the hinges **10**. This means that the second gear **24***b* of the (lower) second bearing device **18***b* is pressed in the direction **M2** by the weight of the door wing **3***a*. The tooth flank **17** of the second gear **24***b* is thus pressed against the teeth of the second toothed rack **19***b* in the direction M2. Due to the co-operation of the oblique tooth flank **17** of the second gear **24***b* with the oblique toothing of the second toothed rack **19***b*, a resulting force presses the second longitudinal side **31***b*, opposing the first longitudinal side **31***a*, of the second toothed rack **19***b* against the second gliding device **30** having the rolling bodies **30***a*. Accordingly, the second gliding device **30** is thus provided to reduce the frictional resistance between the second longitudinal side **31***b* of the second toothed rack **19***b* and the (lower) second bearing device **18***b*.

1. A guide system for guiding at least one movably-supported door wing, in particular a folding-sliding-door, along a furniture wall, the guide system comprising:
at least one carrier for pivotally supporting the at least one door wing,
at least one guide, in particular a guide rail, configured to be fixed to the furniture wall for moving the at least one carrier along the furniture wall,
at least one guide device for displaceably supporting the at least one carrier along the at least one guide, and
at least one toothed rack and at least one gear engaged or configured to be engaged in the toothed rack, the gear being movable at least partially along the toothed rack upon a movement of the carrier along the guide and being configured to be set into rotation by an engagement in the toothed rack,
wherein the at least one carrier includes at least one bearing device for supporting the at least one toothed rack.

2. The guide system according to claim 1, wherein the at least one bearing device and the at least one toothed rack are movable relative to each other, preferably displaceable relative to each other, and/or wherein the at least one bearing device includes at least one linear guide, preferably a U-shaped profile, for movably, preferably displaceably, supporting the at least one toothed rack.

3. The guide system according to claim 1, wherein the at least one toothed rack has an end section, and at least one holding device to be fixed to the furniture wall is provided on which the end section of the toothed rack is supported.

4. The guide system according to claim 3, wherein the at least one toothed rack and the at least one holding device are movable relative to each other, preferably displaceable relative to each other, preferably wherein the at least one holding device includes at least one guide for movably supporting the at least one toothed rack.

5. The guide system according to claim 3, wherein at least one adjustment device is provided for adjusting a position of the at least one toothed rack relative to the at least one holding device, preferably wherein the at least one adjustment device includes at least one rotatable adjustment wheel, and a position of the at least one toothed rack relative to the at least one holding device is adjustable by rotating the at least one adjustment wheel.

6. The guide system according to claim 3, wherein the at least one holding device is configured so as to hold the at least one toothed rack with a distance to the furniture wall in a fixed condition of the holding device.

7. The guide system according to claim 3, wherein the at least one toothed rack has a first end section and a second end section, wherein the first end section of the toothed rack is supported on a first holding device and the second end section of the toothed rack is supported on a second holding device, wherein each of the holding devices is to be fixed to the furniture wall and the bearing device is movable between the two holding devices.

8. The guide system according to claim 1, wherein the at least one gear is rotationally supported on the at least one bearing device or on the at least one carrier.

9. The guide system according to claim 1, wherein at least one gliding device is provided for reducing a frictional resistance between the at least one bearing device and the at least one toothed rack, preferably wherein the at least one gliding device is arranged on the bearing device.

10. The guide system according claim 9, wherein the at least one toothed rack includes a first longitudinal side having teeth, the first longitudinal side being configured to be engaged with the at least one gear, and further includes a second longitudinal side opposing the first longitudinal side, wherein the at least one gliding device is configured to bear against the second longitudinal side of the at least one toothed rack.

11. The guide system according claim 9, wherein the at least one gliding device includes at least one rolling body rotatable about an axis, preferably extending vertically in the mounted position, preferably wherein the gliding device includes a plurality of rolling bodies spaced apart from each other in a longitudinal direction of the at least one toothed rack and/or arranged in at least two running planes lying above each other in the mounted position.

12. The guide system according to claim 1, wherein at least a second toothed rack is provided and the carrier includes a second bearing device for supporting the second toothed rack, wherein the second bearing device is spaced apart from the bearing device in a longitudinal direction of the carrier, wherein at least a second gear is provided which engages or which is configured to be engaged in the second toothed rack, the second gear being movable at least partially along the second toothed rack upon a movement of the carrier along the at least one guide and being configured to be set into rotation by an engagement in the second toothed rack.

13. The guide system according claim 12, wherein the guide system includes at least one synchronization shaft for synchronizing a movement of the two gears.

14. The guide system according claim 13, wherein the guide system includes at least one stabilizing device configured to prevent or at least to limit a bulging movement of the synchronization shaft in a direction extending transversely to a longitudinal direction of the synchronization shaft, preferably wherein the at least one stabilizing device:
is arranged in a central region of the synchronization shaft, and/or
includes at least one opening for the passage of the synchronization shaft, and/or
includes a fastening section configured to bear against the carrier and a holding limb for holding the synchronization shaft, the holding limb projecting transversely, preferably perpendicular, from the fastening section, wherein the fastening section includes at least one fastening location for fixing to the carrier.

15. The guide system according to claim 12, wherein the arrangement consisting of toothed racks, gears, bearing devices and, if appropriate, synchronization shaft for synchronizing a movement of the gears, collectively form a compensating device for compensating for a tilting moment of the carrier or of the door wing arranged thereon about a tilting axis by a return moment.

16. The guide system according to claim 1, wherein the guide system includes at least a second guide, preferably a guide rail, to be fixed to the furniture wall with a vertical distance to the at least one guide, wherein at least one further guide device is provided for displaceably supporting the at least one carrier along the second guide.

17. The guide system according to claim 1, wherein the guide device includes at least one running wheel, preferably a plurality of running wheels, wherein the at least one running wheel is displaceable along the at least one guide,

preferably wherein a plurality of running wheels is rotationally supported about a rotational axis oriented in a first direction, and further running wheels are rotationally supported about a rotational axis oriented in a second direction.

18. An item of furniture comprising a furniture carcass, at least one door wing, in particular a folding-sliding-door, movably supported relative to the furniture carcass, and at least one guide system according to claim **1** for moving the at least one door wing relative to the furniture carcass, preferably wherein the furniture carcass includes at least one furniture wall in the form of a sidewall, wherein the at least one guide, in particular a guide rail, is fixed to the sidewall and wherein the at least one door wing is movable by the guide system along the sidewall of the furniture carcass.

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